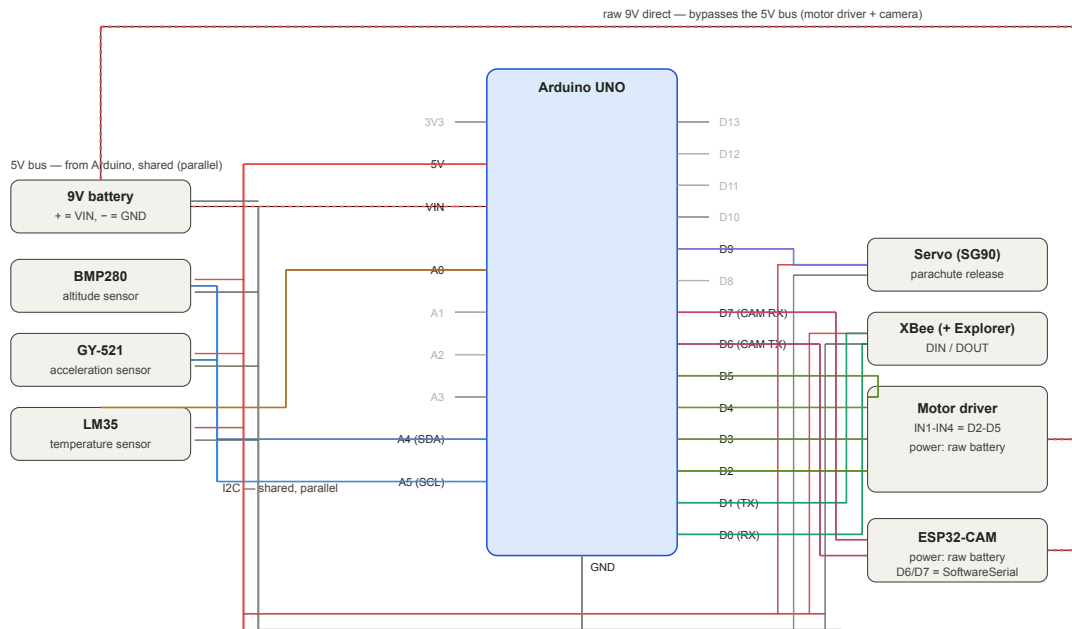




Legend

- Solid red = 5V bus (shared, parallel) - Dashed red = raw 9V direct from battery
- Gray = GND — common return, shared by everything
- Blue = I2C bus (BMP280 + GY-521 share A4/A5, in parallel)
- Orange = LM35 analog signal (A0, dedicated wire)
- Purple = servo signal (D9, dedicated wire)

- Teal = XBee TX/RX (D0/D1 — disconnect before uploading code)
 - Green = motor driver control (D2–D5, dedicated wires)
 - Magenta = ESP32-CAM serial (D6/D7 via SoftwareSerial — kept off D0/D1 to avoid conflict with the XBee)
- Gray pin labels (3V3, A1–A3, D8, D10–D13) = unused pins, no wire connected

Lines that cross without a junction dot are NOT connected — different nets simply pass over each other.

Motor driver and ESP32-CAM are powered directly from the battery (raw 9V) to avoid current spikes disturbing the sensor readings.

All grounds (GND) are common across the whole circuit — battery, Arduino, and every component share the same return path.